

CS 5200 Database Management Systems

Lab 4: Connecting to MySQL instance

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A. Connecting to MySQL instance

- There are many different ways to connect MySQL instance running on GCP that you created before. Among them, you are going to test the following methods:

1. Connecting MySQL instance from your PC using IP Addresses
2. Connecting MySQL Client from Compute Engine
3. Connecting from App Engine
4. Connecting to Cloud SQL from External Applications

- We will practice for the first method here, but you can use other connecting methods using the following instructions:

<https://cloud.google.com/sql/docs/mysql/how-to>

Reminder: How to Get To MySQL Instance

Go to console.cloud.google.com and log in.

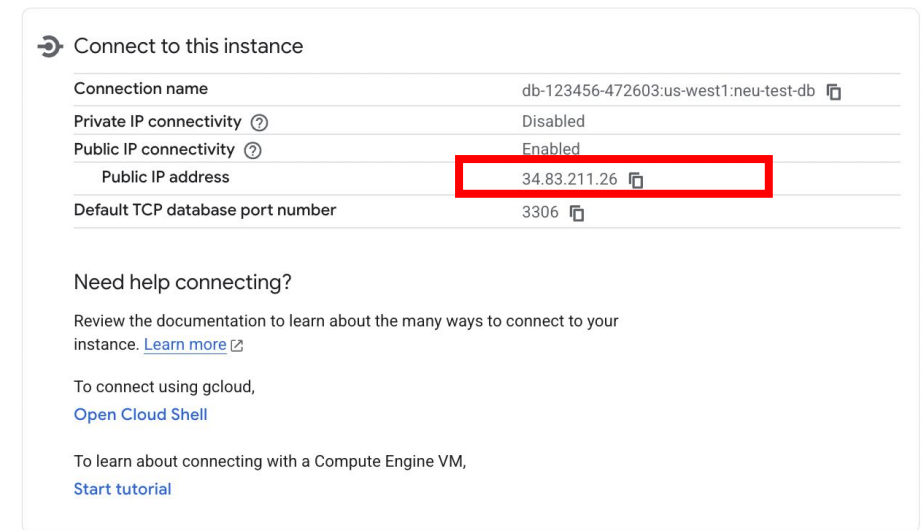
Click the menu icon in top left corner, select Cloud SQL > Instances.

Click into your instance you created last lab, and start it.

Configure access to your MySQL instance

1. Click the instance to open its Overview page, and record its IPv4 address (scroll down to “Connect to this instance”).
 - Note down what the Public IP address of this instance is.
 - We will refer to this as <CLOUD_IPV4>
2. Get your local IPv4 Address (your personal computer’s)
 - If you are not sure how to get it, click <https://ip4.me/>
 - We will refer to this as <LOCAL_IPV4>

Note: You need to re-check and update addresses every time you connect to mysql, as it changes.



Connect to this instance	
Connection name	db-123456-472603:us-west1:neu-test-db
Private IP connectivity	Disabled
Public IP connectivity	Enabled
Public IP address	34.83.211.26
Default TCP database port number	3306

Need help connecting?

Review the documentation to learn about the many ways to connect to your instance. [Learn more](#)

To connect using gcloud, [Open Cloud Shell](#)

To learn about connecting with a Compute Engine VM, [Start tutorial](#)

3. Go to MySQL instance > Connections menu > Networking tab. Check “Public IP” if not checked off already.
4. Create a name (like “MyDevice”) and enter your **<LOCAL_IPV4>** address into “Network”
5. Click **Done**, then click **Save** at the bottom of your page to save your changes

The screenshot shows the Google Cloud console interface for a Cloud SQL instance named 'neu-test-db'. The left sidebar contains a navigation menu with options: Overview, Cloud SQL Studio, System insights, Query insights, **Connections** (highlighted), Users, Databases, Backups, Replicas, and Operations. The main content area is titled 'Connections' and shows the instance 'neu-test-db' with 'MySQL 8.0'. Below this, there are tabs for 'Summary', 'Networking' (selected), 'Security', and 'Connectivity Tests'. The 'Networking' tab contains instructions on how to connect to the instance and a section for 'Instance IP assignment'. In this section, the 'Public IP' option is selected with a checked checkbox, while 'Private IP' is unselected. Below this, there is a section for 'Authorized networks' with a 'Learn more' link. At the bottom of the page, there is a 'Release Notes' link and a 'Google Cloud services authorization' section. A 'Done' button is visible at the bottom right of the 'Networking' tab content.

Connect to your MySQL instance

1. Windows:

- Start cmd.exe on Windows
- Go to where you extract mysql downloaded file on
- **C:/Program Files/MySQL/MySQL Workbench 6.3 CE**
(This varies based on your path selection during installation)

2. MacOS:

- Start Terminal on MacOS

3. Start the mysql client with IP address for MySQL instance

```
$ mysql --host=<CLOUD_IPV4> --user=root --password
```

4. Enter your password

5. You should see mysql prompt

IP address or hostname of your MySQL instance on GCP
If you are connecting to MySQL on your PC, remove this

Connected to MySQL on GCP

```
[Eunjis-MacBook-Pro-3:ws1 eunjilee$ mysql --host=34.83.211.26 -u root -p
```

```
[Enter password:
```

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

```
Your MySQL connection id is 936
```

```
Server version: 8.0.41-google (Google)
```

```
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```

```
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owners.
```

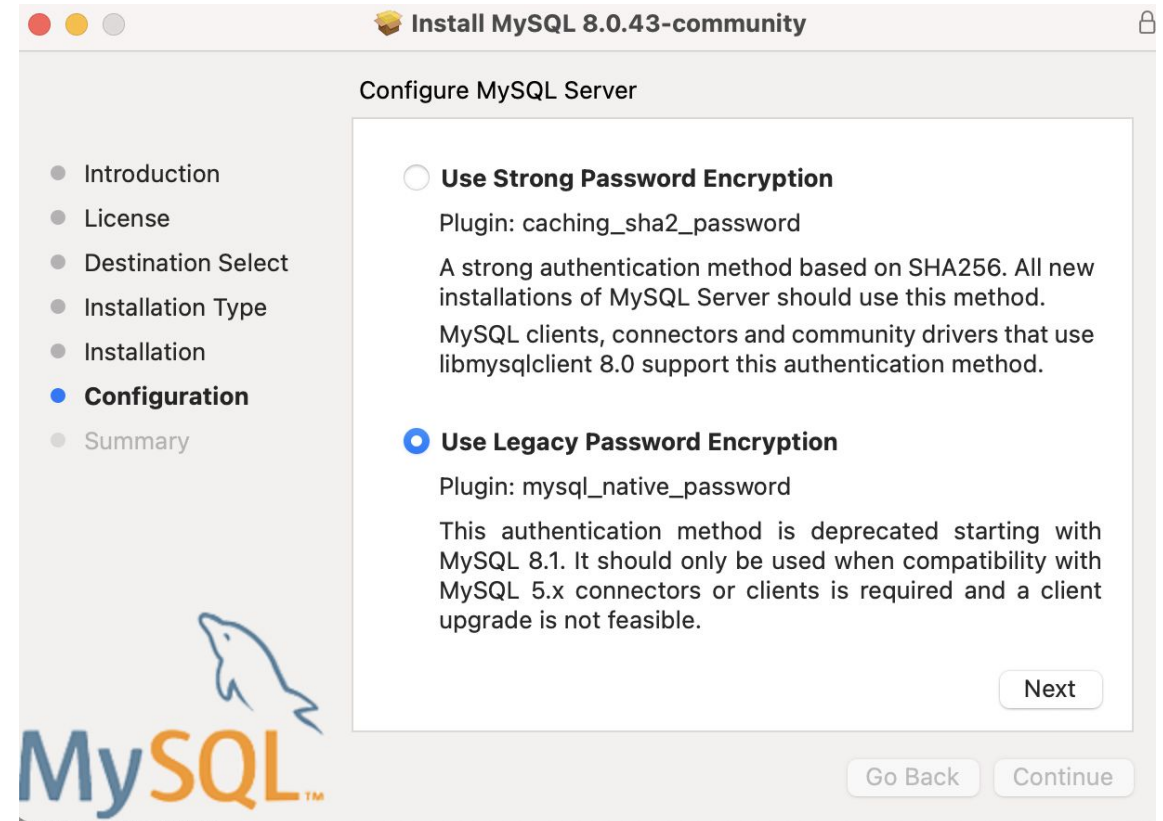
```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> █
```

Common Issues

*Authentication plugin 'mysql_native_password' cannot be loaded:
dlopen(/usr/local/Cellar/mysql/9.4.0_3/lib/plugin/mysql_native_password.so, 0x0002): tried:
'/usr/local/Cellar/mysql/9.4.0_3/lib/plugin/mysql_native_password.so' (no such file),
'/System/Volumes/Preboot/Cryptexes/OS/usr/local/Cellar/mysql/9.4.0_3/lib/plugin/mysql_native_password.so' (no such file),
'/usr/local/Cellar/mysql/9.4.0_3/lib/plugin/mysql_native_password.so' (no such file)*

This means you have MySQL 9.X installed (which uses newer authentication). GCP only supports up to 8.X. Please uninstall and reinstall MySQL with version 8.0, and when you get to this step in the configuration, select “Legacy Password Encryption”



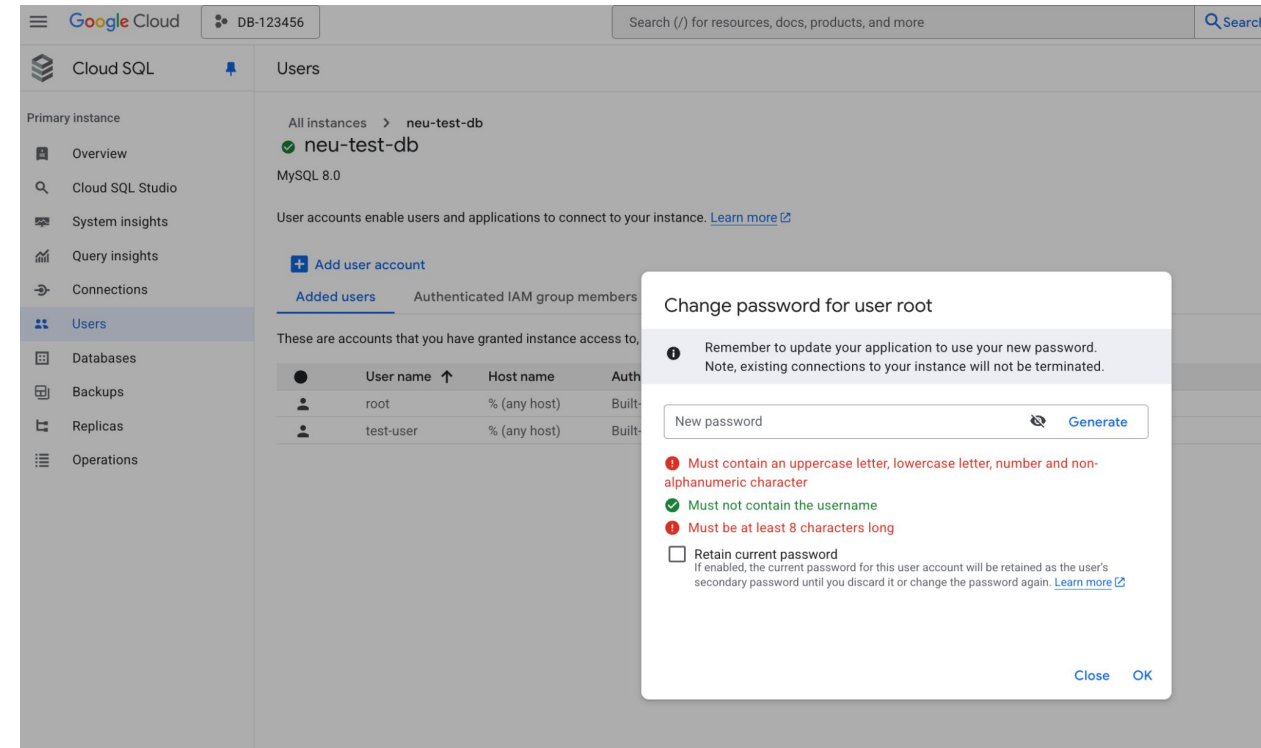
Common Issues

```
Eunjis-MacBook-Pro-3:ws1 eunjilee$ mysql --host=34.83.211.26 -u root -p
```

Enter password:

```
ERROR 1045 (28000): Access denied for user 'root'@'107.3.141.56' (using password: YES)
```

You are likely not using the password for your GCP instance. You noted it down last class when we made the instance; if you lost it, you will have to reset it (under users > root > change password).



B. Taste mysql client (Command Line Tool)

1. help
2. show
3. Useful mysql commands

B1. help

```
mysql> help
```

```
lab3 — mysql -uroot -p — 79x44

[mysql> help

For information about MySQL products and services, visit:
  http://www.mysql.com/
For developer information, including the MySQL Reference Manual, visit:
  http://dev.mysql.com/
To buy MySQL Enterprise support, training, or other products, visit:
  https://shop.mysql.com/

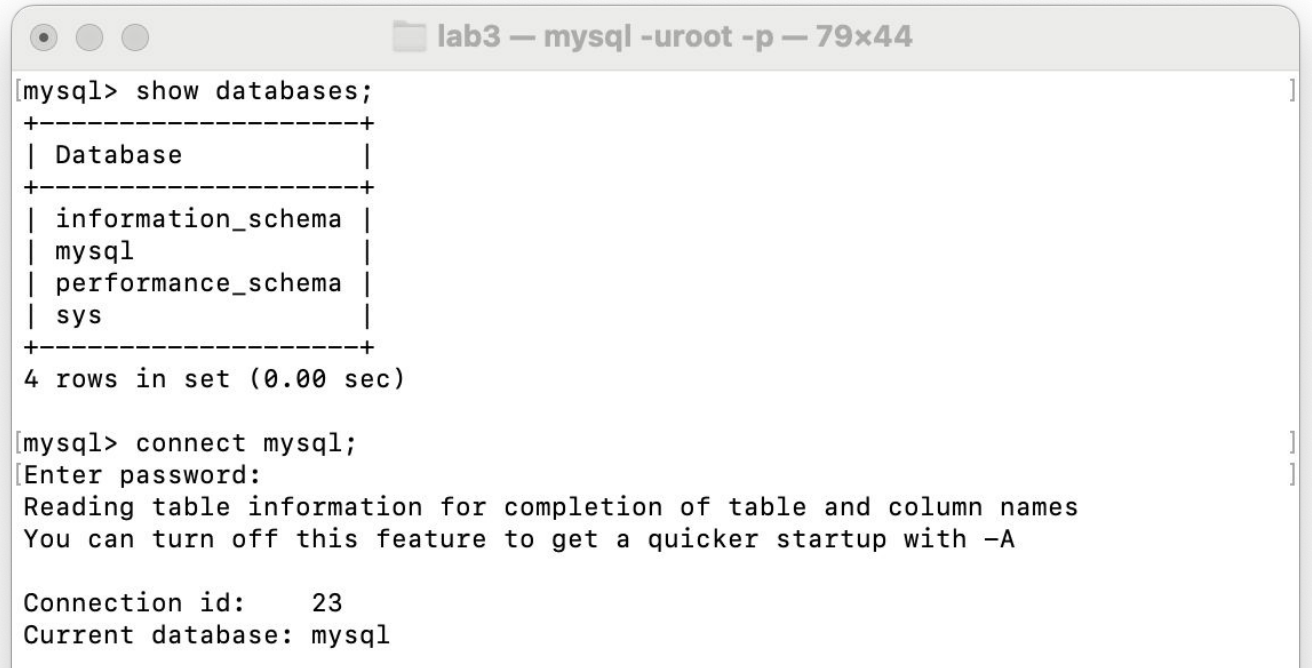
List of all MySQL commands:
Note that all text commands must be first on line and end with ';'
?      (\?) Synonym for 'help'.
clear  (\c) Clear the current input statement.
connect (\r) Reconnect to the server. Optional arguments are db and host.
delimiter (\d) Set statement delimiter.
edit   (\e) Edit command with $EDITOR.
ego    (\G) Send command to mysql server, display result vertically.
exit   (\q) Exit mysql. Same as quit.
go      (\g) Send command to mysql server.
help   (\h) Display this help.
nopager (\n) Disable pager, print to stdout.
notee   (\t) Don't write into outfile.
pager  (\P) Set PAGER [to_pager]. Print the query results via PAGER.
print  (\p) Print current command.
prompt (\R) Change your mysql prompt.
quit   (\q) Quit mysql.
rehash (\#) Rebuild completion hash.
source (\.) Execute an SQL script file. Takes a file name as an argument.
status (\s) Get status information from the server.
system (\!) Execute a system shell command.
tee     (\T) Set outfile [to_outfile]. Append everything into given outfile.
use     (\u) Use another database. Takes database name as argument.
charset (\C) Switch to another charset. Might be needed for processing binlog
        with multi-byte charsets.
warnings (\W) Show warnings after every statement.
nowarning (\w) Don't show warnings after every statement.
resetconnection (\x) Clean session context.
query_attributes Sets string parameters (name1 value1 name2 value2 ...) for the
        next query to pick up.

For server side help, type 'help contents'

mysql> 
```

```
mysql> show databases;
```

```
mysql> connect mysql;
```



```
[mysql> show databases;
+-----+
| Database           |
+-----+
| information_schema |
| mysql               |
| performance_schema |
| sys                 |
+-----+
4 rows in set (0.00 sec)

[mysql> connect mysql;
[Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Connection id:      23
Current database: mysql
```

Note: to run SQL script (.sql file) on mysql client, you need to log in mysql at the same location.
So, you can change a directory (cd) on your terminal, then log in.

```
mysql> help logs;
```

```
mysql> help show binary logs;
```

```
mysql> show binary logs;
```

```
lab3 — mysql -uroot -p — 79x44

Query OK, 0 rows affected (0.00 sec)

[mysql>
[mysql> help logs;
Many help items for your request exist.
To make a more specific request, please type 'help <item>',
where <item> is one of the following
topics:
    PURGE BINARY LOGS
    SHOW
    SHOW BINARY LOGS

[mysql> help show binary logs;
Name: 'SHOW BINARY LOGS'
Description:
Syntax:
SHOW BINARY LOGS
SHOW MASTER LOGS

Lists the binary log files on the server. This statement is used as
part of the procedure described in [HELP PURGE BINARY LOGS], that shows
how to determine which logs can be purged.

mysql> SHOW BINARY LOGS;
+-----+-----+
| Log_name          | File_size |
+-----+-----+
| binlog.000015     | 724935    |
| binlog.000016     | 733481    |
+-----+-----+

URL: http://dev.mysql.com/doc/refman/5.5/en/show-binary-logs.html

[mysql> show binary logs;
+-----+-----+-----+
| Log_name          | File_size | Encrypted |
+-----+-----+-----+
| binlog.000001     | 1810677   | No        |
+-----+-----+-----+
1 row in set (0.01 sec)

mysql> 
```

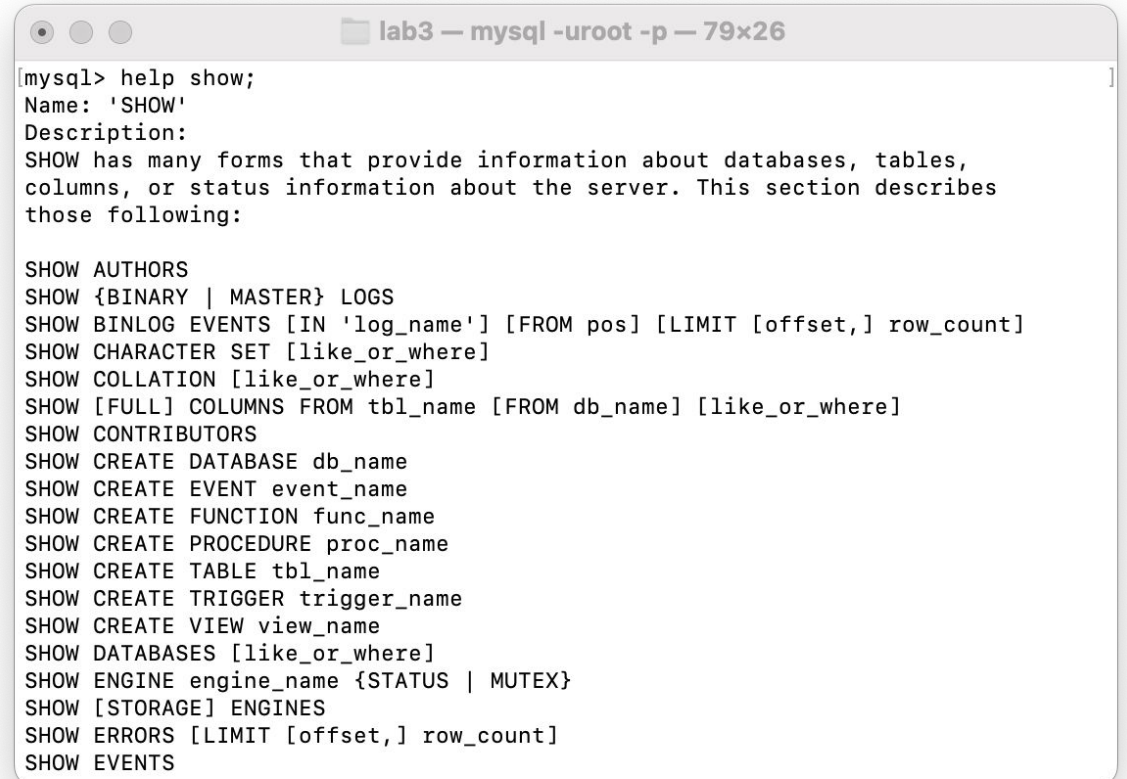
B2. Show

- SHOW has many forms that provide information about databases, tables, columns, or status information about the server. This section describes

```
mysql> help show;
```

```
mysql> show databases;
```

```
mysql> show tables;
```

A screenshot of a MySQL terminal window titled "lab3 — mysql -uroot -p — 79x26". The terminal shows the command "mysql> help show;" and its output. The output includes the command name "SHOW", a description of the command's purpose, and a list of various "SHOW" subcommands such as "SHOW AUTHORS", "SHOW DATABASES", "SHOW TABLES", etc.

```
lab3 — mysql -uroot -p — 79x26
mysql> help show;
Name: 'SHOW'
Description:
SHOW has many forms that provide information about databases, tables,
columns, or status information about the server. This section describes
those following:

SHOW AUTHORS
SHOW {BINARY | MASTER} LOGS
SHOW BINLOG EVENTS [IN 'log_name'] [FROM pos] [LIMIT [offset,] row_count]
SHOW CHARACTER SET [like_or_where]
SHOW COLLATION [like_or_where]
SHOW [FULL] COLUMNS FROM tbl_name [FROM db_name] [like_or_where]
SHOW CONTRIBUTORS
SHOW CREATE DATABASE db_name
SHOW CREATE EVENT event_name
SHOW CREATE FUNCTION func_name
SHOW CREATE PROCEDURE proc_name
SHOW CREATE TABLE tbl_name
SHOW CREATE TRIGGER trigger_name
SHOW CREATE VIEW view_name
SHOW DATABASES [like_or_where]
SHOW ENGINE engine_name {STATUS | MUTEX}
SHOW [STORAGE] ENGINES
SHOW ERRORS [LIMIT [offset,] row_count]
SHOW EVENTS
```

B3. Useful mysql commands

Switch to a database.

```
mysql> use [db name];
```

To see table's field formats.

```
mysql> describe [table name];
```

To return columns and column information.

```
mysql> show columns from [table name];
```

B4. Creating a new user (MySQL 8.X)

Creating a new user. Login as root.

```
mysql> CREATE USER 'username' IDENTIFIED BY 'password';  
mysql> grant all privileges on [db_name].* to 'username';
```

For example, create user 'test' with password '1234' on 'test' database

```
mysql> CREATE USER 'test' IDENTIFIED BY '1234';  
mysql> CREATE DATABASE test;  
mysql> grant all privileges on test.* to 'test';
```

Login as a new user. Then, use test database

```
mysql> use test;
```

Change a users password from MySQL prompt.

Set the password. Update privileges.

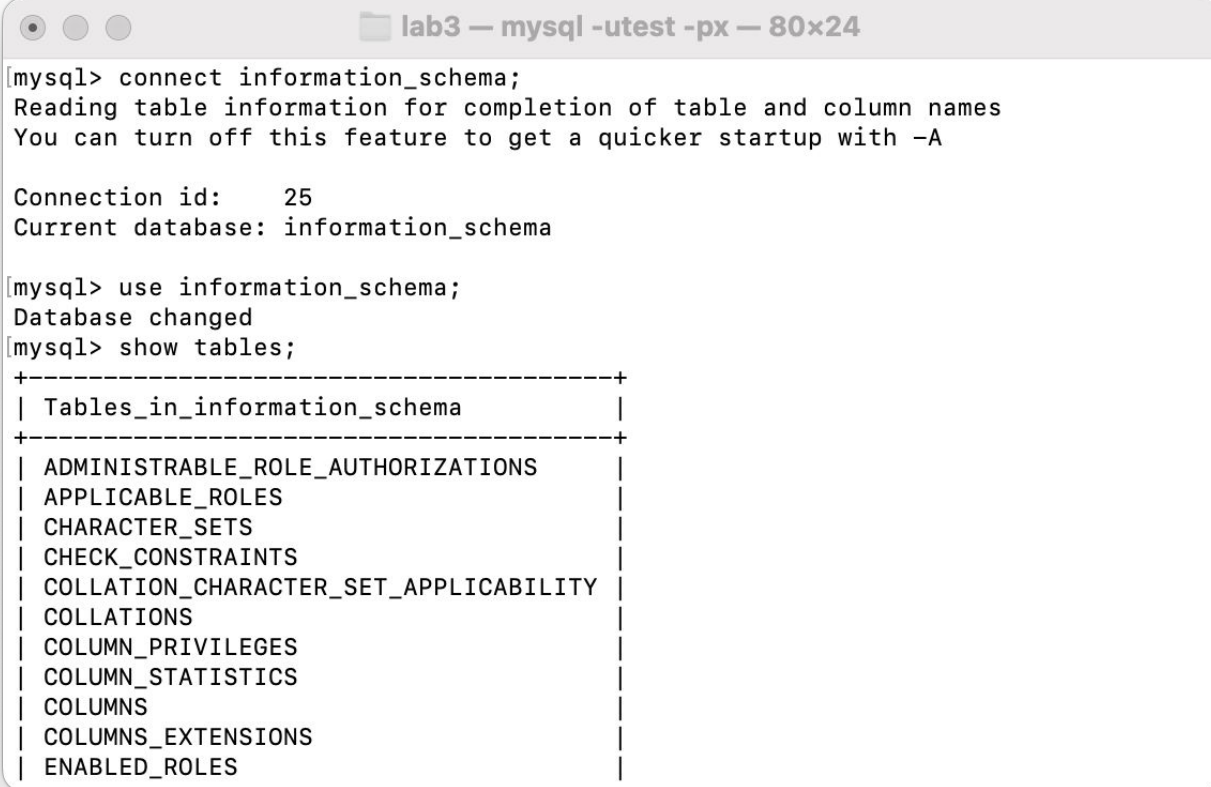
```
mysql> SET PASSWORD FOR 'user'@'hostname' = PASSWORD('password');  
mysql> flush privileges;
```


C. Catalog Database in MySQL (i.e., INFORMATION_SCHEMA)

```
mysql> connect information_schema;
```

```
mysql> use information_schema
```

```
mysql> show tables;
```



```
lab3 — mysql -utest -px — 80x24
[mysql> connect information_schema;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Connection id: 25
Current database: information_schema

[mysql> use information_schema;
Database changed
[mysql> show tables;
+-----+
| Tables_in_information_schema |
+-----+
| ADMINISTRABLE_ROLE_AUTHORIZATIONS |
| APPLICABLE_ROLES |
| CHARACTER_SETS |
| CHECK_CONSTRAINTS |
| COLLATION_CHARACTER_SET_APPLICABILITY |
| COLLATIONS |
| COLUMN_PRIVILEGES |
| COLUMN_STATISTICS |
| COLUMNS |
| COLUMNS_EXTENSIONS |
| ENABLED_ROLES |
```

% Related to TABLE: TABLES

```
mysql> desc TABLES;
mysql> SELECT table_name FROM tables;
```

% Related to INDEX: INNODB_SYS_INDEXES

```
mysql> desc INNODB_SYS_INDEXES;
mysql> SELECT name, table_id, type FROM INNODB_SYS_INDEXES;
```

% Related to COLUMN: COLUMNS

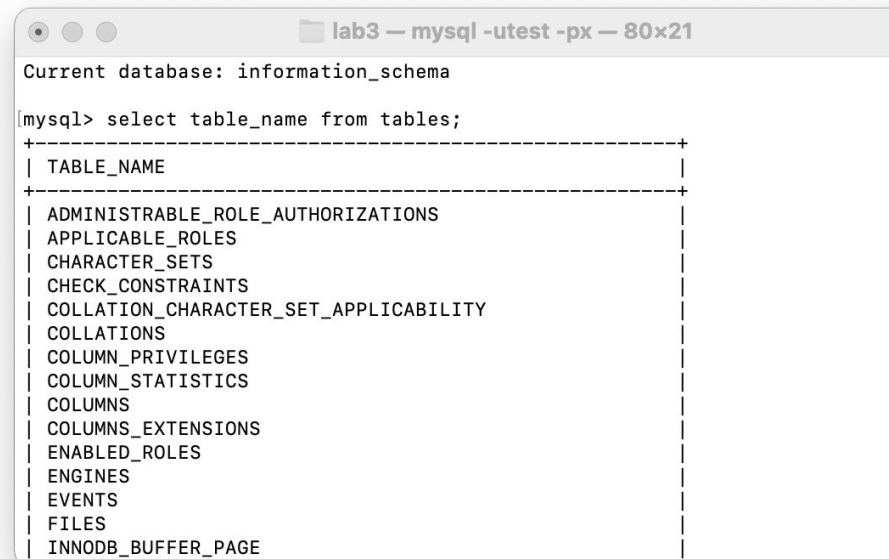
```
mysql> desc COLUMNS;
mysql> SELECT ORDINAL_POSITION, COLUMN_NAME, DATA_TYPE, NUMERIC_PRECISION
-> FROM COLUMNS
-> WHERE TABLE_NAME = 'EMPLOYEE';
```

% Related to VIEWS: VIEWS

```
mysql> desc VIEWS;
mysql> SELECT table_name, VIEW_DEFINITION FROM VIEWS;
```

% Related to CONSTRAINTS: TABLE_CONSTRAINTS

```
mysql> desc TABLE_CONSTRAINTS;
mysql> SELECT constraint_name, table_name, constraint_type
-> FROM TABLE_CONSTRAINTS ;
```



```
lab3 — mysql -utest -px — 80x21
Current database: information_schema

mysql> select table_name from tables;
+-----+
| TABLE_NAME |
+-----+
| ADMINISTRABLE_ROLE_AUTHORIZATIONS |
| APPLICABLE_ROLES |
| CHARACTER_SETS |
| CHECK_CONSTRAINTS |
| COLLATION_CHARACTER_SET_APPLICABILITY |
| COLLATIONS |
| COLUMN_PRIVILEGES |
| COLUMN_STATISTICS |
| COLUMNS |
| COLUMNS_EXTENSIONS |
| ENABLED_ROLES |
| ENGINES |
| EVENTS |
| FILES |
| INNODB_BUFFER_PAGE |
+-----+
```

Create a new Database

% Create a new database “silicon”

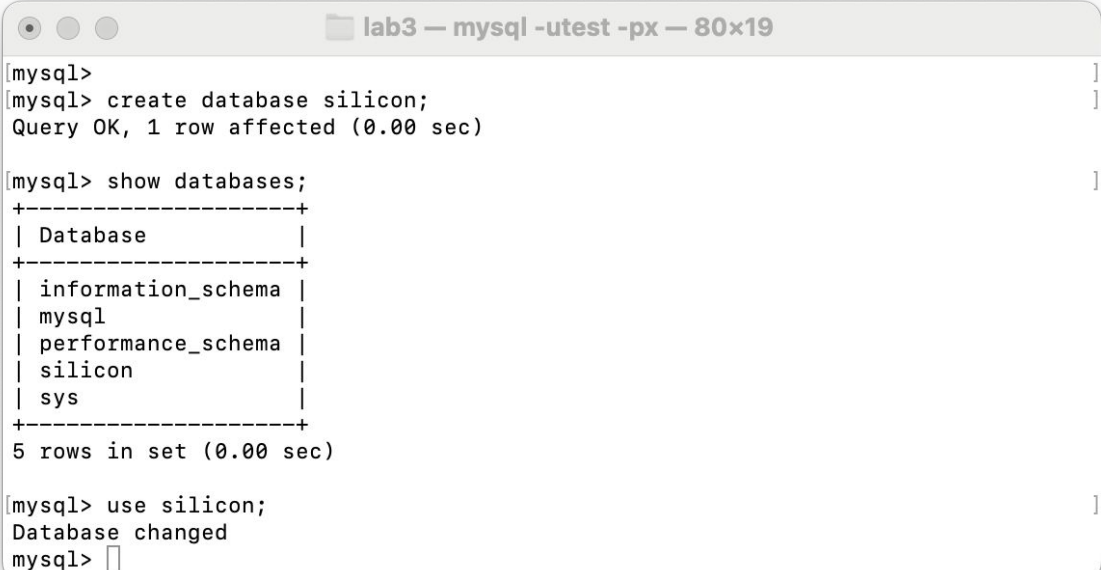
As root user

```
mysql> create database silicon;
```

```
mysql> grant all privileges on silicon.* to 'test-user';
```

As development user (e.g., ‘test-user’ here)

```
mysql> use silicon;
```



```
lab3 — mysql -utest -px — 80x19

[mysql>
[mysql> create database silicon;
Query OK, 1 row affected (0.00 sec)

[mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| silicon |
| sys |
+-----+
5 rows in set (0.00 sec)

[mysql> use silicon;
Database changed
mysql> ]
```

[Lab4 Assignment]

- Please, complete the following and then submit the captured file to Canvas.
 - Capture a screenshot of mysqls that shows success of the following commands (you can copy paste multiple screenshots into a google doc and submit)
 - Connect to MySQL on GCP
 - Create a user
 - Create a database
 - Show databases
- Submit the screen shots into Canvas.

Example Submission

```
Eunjis-MacBook-Pro-3:ws1 eunjilee$ mysql --host=34.83.211.26 -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 936
Server version: 8.0.41-google (Google)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

connect to cloud mysql instance (must show
host = <CLOUD_IPV4>, not your local mysql!

Show database command

Successfully create a new user

Successfully create a new
database

```
mysql> CREATE USER "test-user" IDENTIFIED BY "12345678a!";
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> create database test;
Query OK, 1 row affected (0.03 sec)
```

```
mysql> grant all privileges on test.* to "test-user";
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> exit
```

```
Bye
Eunjis-MacBook-Pro-3:ws1 eunjilee$ mysql --host=34.83.211.26 -u test-user -p
Enter password:
[ERROR 1045 (28000): Access denied for user 'test-user'@'107.3.141.56' (using password: YES)
Eunjis-MacBook-Pro-3:ws1 eunjilee$ mysql --host=34.83.211.26 -u test-user -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 1158
Server version: 8.0.41-google (Google)
```

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> use test;
Database changed
mysql> show tables;
Empty set (0.03 sec)
```

```
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| performance_schema |
| test      |
+-----+
3 rows in set (0.03 sec)

mysql> █
```

Successfully login as your
new user (note that
password is the new one
you set for your new user,
not your root password!)

Don't forget to turn off your
Instance!!!!!!!